

CHAE-YEON HEO

Computer Vision Researcher · Generative Models & 3D Vision

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The primary goal of my research focuses on developing computer vision and generative models that can intelligently understand, edit, and create visual content. I aim to build methods that connect high-level semantic interpretation with practical, controllable image manipulation — covering tasks such as image restoration, inpainting, and generative editing. I am also interested in 3D vision, including reconstruction and neural rendering, with the broader goal of improving the reliability and flexibility of visual AI systems used in real-world applications.

RESEARCH INTERESTS

Computer Vision · Image & Video Editing / Restoration · Generative Models · 3D Vision

EDUCATION

Chonnam National University Sep 2024 – Aug 2026
M.S. in Artificial Intelligence Convergence · Advisor: Prof. Yeong-Jun Cho · GPA 4.31/4.5

Chonnam National University Mar 2021 – Aug 2024
B.S. in Artificial Intelligence · GPA 3.57/4.5

RESEARCH EXPERIENCE

NAVER LABS — Research Intern, Spatial AI Team Sep 2026 – Mar 2027 (Expected)
Contributing to applied research on generative 3D scene reconstruction (3D Gaussian Splatting) for large-scale spatial AI systems. Project scope and technical details are confidential.

TECHNICAL SKILLS

Languages: Python
Libraries & Frameworks: PyTorch, OpenCV, NumPy, Pandas, Matplotlib, HuggingFace
Research Areas: Image & Video Editing / Restoration , Generative Models , 3D Vision (Gaussian Splatting) , Computer Vision
DevOps & Tools: Git, Linux, LaTeX, Docker

PUBLICATIONS

- 1. HG-Delegator: Harmonized Geometry Delegation for Cross-Frame Consistent Video Object Removal** **UNDER REVIEW**
Authors: **Heo, C. Y.**, & Cho, Y. J. · IEEE Transactions on Multimedia (TMM), 2026 — Special Section on “Multimodal Video Compression and Reconstruction: Theory, Algorithms, and Applications” ★ JCR Q1 IF 9.7 TOP 3.1%
Contribution: Proposes a reliability-aware framework that aligns visible background observations from multiple video frames in a shared 3D representation and reprojects them into each target view as a confidence-weighted background prior, achieving cross-frame consistent video object removal and the best PSNR, MS-SSIM, and LPIPS across six benchmarks.
- 2. Anonymous Submission — AAAI 2027** **UNDER REVIEW**
Authors: **Heo, C. Y.**, & Cho, Y. J. · Double-blind review in progress — title and details withheld
Contribution: A progressive large-mask image inpainting framework for diffusion-based restoration.
- 3. CSF-Net: Context-Semantic Fusion Network for Large Mask Inpainting** **PUBLISHED** h5-index 131
Authors: **Heo, C. Y.**, & Cho, Y. J. · WACV 2026 — Winter Conference on Applications of Computer Vision
Contribution: Presents the first inpainting framework that leverages amodal completion as semantic guidance for large-mask image restoration, using a transformer-based fusion strategy that unifies amodal priors with contextual cues to substantially reduce object hallucination.
- 4. Flatfish Lesion Detection Based on Part Segmentation Approach and Disease Image Generation** ★ TOP 10% SCI JCR Q1 IF 4.7
Authors: Hwang, S. B., Kim, H. Y., **Heo, C. Y.**, Jeong, H. Y., Jung, S. J., & Cho, Y. J. · Journal of the World Aquaculture Society, 2025 — JCR percentile 9.7%
Contribution: Introduces the first high-resolution, expert-annotated flatfish lesion dataset and a part-wise lesion detection framework, with a GAN-based disease image synthesis and harmonization pipeline that improves detection accuracy by up to 12%.
- 5. Missing Person Recognition Algorithms Based on Image Captioning and Visual Grounding** **PUBLISHED** h5-index 68
Authors: Jeong, A., Woo, Y., Kim, H. Y., Suh, G., **Heo, C. Y.**, Cho, Y. J., & Jeong, H. · ICPR 2024 — International Conference on Pattern Recognition
Contribution: Proposes a multimodal missing-person recognition pipeline that jointly uses image captioning and visual grounding for fine-grained attribute reasoning under incomplete or ambiguous visual evidence.

Authors: Seo, Y. R., Heo, C. Y., & Kim, C. S. · Journal of Digital Contents Society, 2024

Contribution: Introduces the first sensor-fusion 3D Gaussian Splatting pipeline integrating LiDAR-SLAM pose alignment and LiDAR depth supervision, enabling physically accurate real-scale scene reconstruction for autonomous-driving simulation.

PATENTS & SOFTWARE

Patents

- 3D Drone Position Estimation Using a Single Camera — KR Patent Registered (10-2025-0089527)
- Image Restoration System Based on a Context-Semantic Fusion Network for Large-Scale Mask Inpainting — KR Patent Registered (10-2026-0020469)

Software

- Camera–LiDAR Sensor Fusion–based Real-Scale Environment Rendering — Software Registered (C-2025-027288), Jul 2025
- 3D Drone Trajectory Estimation in Monocular Camera Environments — Software Registered (C-2025-017676), May 2025
- Flatfish Lesion Detection System Based on Three-Region Fish Segmentation — Software Registered (C-2025-060751), Dec 2025
- Image Restoration System Based on a Context-Semantic Fusion Network for Large-Scale Mask Inpainting — Software Registered (C-2025-060749), Dec 2025

SELECTED PROJECTS

Industrial Safety Anomaly Detection System (Individual Project)	Jul 2026
Built a real-time industrial-site safety monitoring system that detects worker falls/slips and violations (PPE non-compliance, unauthorized zone entry) via skeleton-based action recognition and fine-tuned object detection, issuing natural-language event alerts.	
Sleep Video-based Airway Obstruction Analysis Model (Individual Project)	Mar 2025 – Dec 2025
Developed an AI system for analyzing Drug-Induced Sleep Endoscopy (DISE) videos using ResNet50 for anatomical classification and VQA, with a labeling-free automated data pipeline for airway tracking and clinical insight generation.	
AiMemory: AI-based Parenting Record Automation Service (Team Lead)	Nov 2025 – Present
Designed and prototyped a multimodal AI system that automates baby diaries, generates one-click photobooks, and ensures on-device privacy to reduce parents’ record-keeping burden.	
Single Image Reflection Removal (Samsung Electronics Joint Project)	Sep 2024 – Jan 2025
Developed a cross-attention–based reflection removal model to fix reflection-induced anomaly detection failures inside washing-machine monitoring systems.	
Jejueo–Standard Translator	Mar 2024 – Jun 2024
Built a bidirectional translation app between Jejueo dialect and Standard Korean to support tourism and dialect preservation.	
Static Environment Modeling using Camera–LiDAR Sensor Fusion	Mar 2024 – Jun 2024
Developed a sensor-fusion pipeline to reconstruct static 3D environments by integrating camera and LiDAR data.	
Nighttime Object Detection using Sensor Fusion	Mar 2024 – Jun 2024
Performed camera–LiDAR calibration and data acquisition using mobile robots, and developed a nighttime object detection pipeline.	
Missing Person Recognition (Team Lead)	Oct 2023 – Dec 2023
Built a recognition pipeline that searches for missing persons using textual appearance descriptions, combining image captioning and visual grounding.	
Zero-shot Labeling Tool using Grounded-SAM (Individual Project)	Nov 2023 – Dec 2023
Created a zero-shot labeling tool leveraging Grounded-SAM to automate dataset annotation for downstream tasks.	
Building Coverage Ratio Measurement via Aerial Image Segmentation	May 2023 – Jul 2023
Implemented segmentation-based measurement of building coverage ratio using aerial imagery.	
CloseCV: Privacy-Preserving Video Re-Identification Solution	Feb 2023
Developed a selective de-identification system using YOLO, ByteTrack, and Track ID–based identity removal; deployed as a competition prototype.	

AWARDS

Silver Prize , 2025 Industry–Academia AI Hackathon Competition (CNU) VLM-based Wildfire Risk Detection System	Jul 2025
Grand Prize , 2024 Capstone Design Excellence Competition (CNU) Sensor Fusion–based Static Environment Modeling	Aug 2024
Gold Prize , 2024 AICOSS Winter Industry–Academia AI Hackathon Missing Person Recognition algorithms based on Image Captioning	Jan 2024

Gold Prize , 2023 AICOSS Winter Industry–Academia AI Hackathon <i>Privacy-Preserving Service Using Object Detection</i>	<i>Jan 2023</i>
Excellence Award , 2022 Winter Energy+AI Micro Capstone Competition (CNU) <i>YOLOv7-based Automated PET Bottle Sorting System</i>	<i>Nov 2022</i>
Grand Prize , 2022 Summer Energy+AI Micro Capstone Competition (CNU) <i>Real-Time Detection of Illegal Filming Drones Using Deep Learning</i>	<i>Aug 2022</i>
Encouragement Award , 2022 Gwangju AI Problem-Solving Idea Contest <i>AI-Based Health Exercise Assistant “A-Gym”</i>	<i>Jul 2022</i>

TEACHING & MENTORING EXPERIENCE

Teaching Assistant — Industry–Academia Capstone Design (Prof. Young-Jun Cho)	<i>Sep 2025 – Dec 2025</i>
Teaching Assistant — AI/Software Capstone Design (Prof. Young-Jun Cho)	<i>Mar 2025 – Aug 2025</i>
Instruction Support — Gwangju High School	<i>Dec 2024</i>
Mentor — Gwangju High School AI Project Program	<i>Jun 2024 – Dec 2024</i>
Mentor — Undergraduate–Graduate Joint Program	<i>Sep 2024 – Jan 2025</i>
Teaching Assistant — Computer Vision (Prof. Young-Jun Cho)	<i>Mar 2024 – Jul 2024</i>
Mentor — SW Teaching Excellence Program (CNU Software-Centered University)	<i>Mar 2024 – Jul 2024</i>
Teaching Assistant — Object-Oriented Design Project (Prof. Young-Jun Cho)	<i>Sep 2023 – Dec 2023</i>
Teaching Assistant — SW/AI Gifted Education Program (CNU Gifted Center)	<i>Mar 2023 – Aug 2023</i>